Cultural Homophily in Grand Slam Doubles

Part 1: ATP Men's Grand Slam Doubles only, 2018–2025 (Olympics excluded throughout — reported separately in Part 2 once this analysis is finalized) · Regression sample: 3,728 team-obs (complete ranking, nationality/language for all 4 players; years since turning pro imputed to 1 for rookies)

1. Data and Measures

Three culture measures capture teammate similarity, all binary (0/1), built from the CEPII gravity dataset's country-pair language indicators. **Same nationality** is shared citizenship. **Same official language** is CEPII's `comlang_off` (shared official/ national language). **Language proximity** is CEPII's `comlang_ethno` (shared widely-spoken ethnic/native language — a language spoken as a mother tongue by at least 9% of the population in both countries), confirmed binary directly against the raw data (only {0,1} values present). It differs from same-language only in using ethnic/spoken-language overlap rather than official designation, so the two can diverge when a country's official language differs from its most widely spoken native language. Shares reported elsewhere in this report (e.g. "56.8%") are the *share* of observations with ling_prox=1, not an average distance — this is not a continuous closeness score.

Regressions use a team-level panel with tournament-by-year fixed effects, round fixed effects, and standard errors clustered by match. Controls in all main specifications: team average doubles ranking, opponent average ranking, top-100 singles indicator (either player), team average years since turning pro, demeaned (*exp_mean_dm*), and its square (*exp_mean_dm_sq*). Culture measures enter one at a time. Rank gap between teammates was tested as an additional control and is insignificant throughout; it is excluded from all main specifications. Nationality and language data are fully resolved in the pipeline (birthplace fallback, surname lookup, Monaco language correction, and `manual_nationality.csv`); zero matches are dropped for nationality.

2. Observation Breakdown

Step	Description	Matches	Change	Team-obs
1	Raw scraped Grand Slam matches (2018–2025, incl. AO/RG/USO 2020; Wimbledon 2020 cancelled)	1,933	—	3,866
2	Drop: qualifying-round matches (found 2026-09-03: Wimbledon 2018 alone contains 12 qualifying-round doubles matches — 8 "1st Round Qualifying" + 4 "2nd Round Qualifying" — not present for any other tournament-year in this dataset; a data-inclusion inconsistency, not a real difference in draw size)	1,921	−12	3,842
3	Drop retirements / walkovers (19 walkovers · 10 retired in set 1 · 18 retired in set 2+)	1,874	−47	3,748
4	Drop: doubles ranking incomplete for ≥1 player (4 of the original 14 were retrieved and restored — see note below)	1,864	−10	3,728
5	Drop: nationality / language missing for ≥1 player (pipeline fully resolves: birthplace, surname lookup, Monaco fix, manual_nationality.csv)	1,864	0	3,728
6	Years since turning pro: 216 obs are rookies (turned pro in/after the tournament year); *exp_mean* imputed to 1 — no obs dropped (these teams have mean rank ≈211 vs ≈121 for the rest; dropping would create selection bias)	—	0	3,728
→ Grand Slams regression sample (Tables 3–4)		**1,864**		**3,728**

Of the 14 matches originally dropped for incomplete doubles ranking, 4 were retrieved: 3 (match_id 6, 35, 49) failed only because Guillermo García-López's accented surname didn't match the unaccented scrape during the ranking merge — his doubles rank was matched in manually; 1 (match_id 913, 2021 Wimbledon) had a genuinely blank partner slot in the raw scrape (Alejandro Davidovich Fokina) — his full profile (rank, nationality, tenure, top-100 history) was reconstructed from public sources and the project's own gravity/ language lookup table. The remaining 10 could not be reliably retrieved and stay dropped.

Table 4 (tiebreak win): 1,177 standard 7-pt tiebreaks × 2 = 2,354 team-obs (923 unique matches), covering sets 1–5. Advantage-set deciders with no breaker played (e.g. 8–6) are not tiebreaks and are excluded entirely — confirmed by the raw tiebreak-score columns being null for all 15 such cases. No genuine 10-point super-tiebreak exists anywhere in this dataset, so there is currently no valid robustness spec to add.

3. Pressure Outcome Counts

Computed on the **same 1,864-match regression sample** used in Tables 3–6 (not a broader pre-filter set), so every number below reconciles exactly with the regression tables.

Outcome	N	Share of matches
Set-1 tiebreak (7-pt)	459	24.6%
Set-2 tiebreak (7-pt)	471	25.3%
Set-3 tiebreak (7-pt)	210	11.3%
Set-4 tiebreak (7-pt)	31	1.7%
Set-5 tiebreak (7-pt)	6	0.3%

Grand Slams regression sample (N=1,864). Sum of the five set-by-set rows: 459 + 471 + 210 + 31 + 6 = **1,177** — this matches Table 4's tiebreak count exactly, since both are computed on the identical sample. Advantage-set deciders with no breaker played (e.g. 8–6, 16–14) are excluded entirely from every row above — a real breaker was never played in those cases, confirmed by the raw tiebreak-score columns being null.

3.1 Match Format: Best-of-3 vs. Best-of-5

Wimbledon played men's doubles as **best-of-5 sets in 2018, 2019, 2021, and 2022** (253 matches played under that rule within the regression sample, of which 101 actually required a 4th or 5th set — the rest finished in straight sets). Every other tournament-year in the Grand Slams — Australian Open, Roland Garros, and US Open in every year, and Wimbledon from 2023 onward — is best-of-3, with no match ever recording a 4th set.

Tournament	Years (best-of-5)	Matches (best-of-5)	Reached 4 sets	Reached 5 sets
Wimbledon	2018, 2019, 2021, 2022	253	101	50
All other tournament-years	—	Best-of-3 throughout	0	0

4. Regression Results

Average marginal effects (AME, dP/dx) from logit models. Each column is a separate model. Standard errors in parentheses, clustered by match. FE: tournament × year, round. All specifications include years-since-turning-pro controls (*exp_mean_dm*, demeaned, and its square). Culture measures enter one at a time.

Table 3 — Outcome: Match Win | N = 3,728 team-obs | AME (dP/dx)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0361**	+0.0459***	+0.0419***
	(0.0164) p=0.028	(0.0157) p=0.003	(0.0157) p=0.008
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	−0.0042	−0.0030	−0.0024
	(0.0315) p=0.894	(0.0315) p=0.925	(0.0315) p=0.940
Team avg. yrs since turning pro	+0.0044***	+0.0043**	+0.0042**
	(0.0017) p=0.008	(0.0017) p=0.010	(0.0017) p=0.011
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0008***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.096	0.097	0.097

*** p<0.01 ** p<0.05 * p<0.10. Clustered SE. FE: tournament×year, round. All three culture measures statistically significant at the 5% level (nationality p=0.028, language p=0.003, proximity p=0.008). exp_mean is demeaned (exp_mean_dm = exp_mean – sample mean of 11.93

years since turning pro) and its square is built from the demeaned variable (exp_mean_dm_sq = exp_mean_dm²), applied consistently across every table in this report. This is a pure reparameterization — it leaves the fitted model, the culture AMEs, and every other control's AME unchanged; it only changes what "Team avg. yrs since turning pro" reads as (the marginal effect of tenure evaluated at the sample-mean tenure level rather than at zero tenure). Tenure shows a concave effect: positive linear ($p<0.05$) at the mean, negative quadratic ($p<0.01$). Top-100 singles indicator is not significant: within-GS doubles, having a top singles player does not predict match win once ranking and tenure are controlled.

Table 3d — Robustness: Continuous Linguistic Proximity | N = 3,728 team-obs | AME (dP/dx)

Column (3) of Table 3 measures linguistic proximity as CEPII's binary `comlang_ethno` indicator. As a robustness check, we replace it with three continuous measures, one at a time, same sample/controls/FE as Table 3: **prox1** (Melitz & Toubal 2014, Ethnologue tree-based — the same source Békés & Ottaviano 2025 use for their own language-similarity variable), plus two independent robustness measures from PSW2024 (geopoliticaldistance.org): an inverted tree-distance index and a cognate/lexical-based proximity index. All three are continuous on [0,1] with complete (100%) coverage on this sample, after the same Monaco fallback rule already used for `same_language`/`ling_prox`.

Variable	(1) prox1	(2) PSW tree	(3) PSW cognate
Linguistic proximity (continuous)	$+0.0527^{***}$	$+0.0648^{***}$	$+0.0719^{***}$
	(0.0201) p=0.009	(0.0231) p=0.005	(0.0231) p=0.002
Team avg. doubles rank	-0.0007^{***}	-0.0007^{***}	-0.0007^{***}
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	$+0.0007^{***}$	$+0.0007^{***}$	$+0.0007^{***}$
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	−0.0034	−0.0038	−0.0034
	(0.0314) p=0.914	(0.0314) p=0.905	(0.0315) p=0.914
Team avg. yrs since turning pro	$+0.0044^{***}$	$+0.0046^{***}$	$+0.0047^{***}$
	(0.0017) p=0.009	(0.0017) p=0.006	(0.0017) p=0.005
Yrs since turning pro, squared	-0.0008^{***}	-0.0008^{***}	-0.0008^{***}
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.096	0.097	0.097

*** $p<0.01$ ** $p<0.05$ * $p<0.10$. Clustered SE. FE: tournament×year, round. AME reported per full-scale (0→1) change in the linguistic-proximity measure, directly comparable to Table 3 column (3)'s binary AME (+0.0419). Per-SD AMEs: prox1 = 0.0206, PSW tree = 0.0225, PSW cognate = 0.0249. All three continuous measures agree with the binary specification — positive, significant at the 1% level, and somewhat larger in magnitude (5.3–7.2 percentage points for a full-scale change) — so the Table 3 finding is not an artifact of CEPII's binary threshold rule or specific to one continuous data source.

Table 4 — Outcome: Tiebreak Win | Unit: team × tiebreak | AME (dP/dx)

Standard 7pt tiebreaks only, any set (1–5) — N = 2,354 team-obs (1,177 tiebreaks, 923 unique matches). Advantage-set deciders with no breaker played are not tiebreaks and are excluded; no genuine 10-point super-tiebreak exists anywhere in this GS dataset, so there is currently nothing valid to add as a robustness spec. Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (same as Table 3; exp_mean demeaned against this panel's own estimation-sample mean). FE: tournament×year + round. SE clustered by match.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0308	+0.0082	+0.0026
	(0.0224) p=0.169	(0.0218) p=0.706	(0.0217) p=0.905
Team avg. doubles rank	−0.0003***	−0.0003***	−0.0003***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles rank	+0.0003***	+0.0003***	+0.0003***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player	−0.0675	−0.0646	−0.0644
	(0.0428) p=0.114	(0.0427) p=0.130	(0.0427) p=0.131
Team avg. yrs since turning pro	+0.0027	+0.0023	+0.0022
	(0.0022) p=0.217	(0.0021) p=0.285	(0.0021) p=0.297
Yrs since turning pro, squared	+0.0000	+0.0001	+0.0001
	(0.0003) p=0.915	(0.0003) p=0.850	(0.0003) p=0.838
Pseudo-R^2	0.013	0.013	0.013
N	2,354	2,354	2,354

Unit = one team per tiebreak (2 obs per tiebreak), covering sets 1–5. Culture AMEs are small and insignificant across all three measures — homophily advantage does not extend to within-match tiebreak outcomes. Ranking controls remain strongly significant ($p<0.001$), confirming the model captures real quality differences; tenure controls are not significant in the tiebreak sample.

4.1 Robustness: Age at Tournament vs. Years Since Turning Pro

`age_mean` (team average player age at the tournament, from date of birth) is a second natural proxy for career maturity alongside `exp_mean` (years since turning pro). The two are highly correlated — **corr(age_mean, exp_mean) = 0.83**; a regression of age_mean_dm on exp_mean_dm + exp_mean_dm² gives $R^2 = 0.70$, i.e. an **implied VIF of 3.32** for age if entered alongside experience. Entering both together (Table 3-JOINT below) destabilizes each tenure term's own coefficient — both become statistically insignificant even though each is significant on its own (Table 3 vs. Table 3-AGE) — so age is estimated in its own separate specification rather than added alongside experience in the Table 3/4 main specs. Age coverage is complete (0 missing) in both the match-win and tiebreak-win samples, so no rookie-style imputation is needed for age the way exp_mean requires for 216 rookie team-obs.

Table 3-AGE — Outcome: Match Win, age_mean replacing exp_mean | N = 3,728

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0332**	+0.0447***	+0.0409***
	(0.0164) p=0.043	(0.0158) p=0.005	(0.0157) p=0.009
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
Top-100 singles player	+0.0077	+0.0086	+0.0089
	p=0.805	p=0.784	p=0.776
Team avg. age at tournament	+0.0053**	+0.0052**	+0.0051**
	(0.0023) p=0.022	(0.0023) p=0.024	(0.0023) p=0.028
Age, squared	−0.0005	−0.0006	−0.0005
	p=0.154	p=0.141	p=0.146
Pseudo-R^2	0.094	0.095	0.094

Same sample, FE (tournament×year + round), and clustering (by match) as Table 3. Culture AMEs are essentially unchanged from Table 3 (e.g. same nationality: +0.0361 → +0.0332, both $p<0.05$), confirming the core finding is not an artefact of which career-maturity proxy is used. Age itself is positive and significant, concave (like experience), but the quadratic term is not significant here ($p>0.14$).

Table 3-JOINT — age_mean and exp_mean entered together (diagnostic, not a preferred spec) | N = 3,728

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0376**	+0.0467***	+0.0425***
Team avg. yrs since turning pro	+0.0028	+0.0027	+0.0028
	(0.0029) p=0.344	(0.0029) p=0.350	(0.0029) p=0.331
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0008***
Team avg. age at tournament	+0.0026	+0.0025	+0.0022
	(0.0040) p=0.511	(0.0040) p=0.540	(0.0040) p=0.585
Age, squared	+0.0001	+0.0001	+0.0001

Diagnostic only. With both tenure proxies in the model, neither the linear experience term nor the linear age term remains statistically significant ($p>0.33$ for both, in every culture spec), and their standard errors roughly double relative to their single-proxy specs (e.g. exp_mean_dm SE: 0.0017 → 0.0029). This is the expected signature of near-collinear regressors, not evidence that career maturity stops mattering — Table 3 and Table 3-AGE each show a significant, similarly-sized tenure effect on its own. The culture AMEs themselves are robust throughout (+0.0332 to +0.0467, still significant at 5% or better in every spec across Tables 3, 3-AGE, and 3-JOINT).

Table 4-AGE — Outcome: Tiebreak Win, age_mean replacing exp_mean | N = 2,354

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	+0.0253	+0.0043	−0.0013
	(0.0227) p=0.264	(0.0221) p=0.845	(0.0219) p=0.953
Team avg. doubles rank	−0.0003***	−0.0003***	−0.0003***
Opponent avg. doubles rank	+0.0003***	+0.0003***	+0.0003***
Top-100 singles player	−0.0691	−0.0672	−0.0671
	p=0.107	p=0.116	p=0.116
Team avg. age at tournament	−0.0002	−0.0009	−0.0010
	(0.0030) p=0.935	(0.0030) p=0.765	(0.0030) p=0.739
Age, squared	+0.0002	+0.0002	+0.0002
Pseudo-R^2	0.013	0.012	0.012

Same tiebreak sample, FE, and clustering as Table 4. As with experience, age is not significantly associated with tiebreak win (p>0.6 throughout), and — as in Table 4 — none of the three culture measures reach significance either. The null tiebreak result is not an artefact of which career-maturity proxy is used.

4.4 Robustness: Double-Counting / Non-Independence Check

Per Lingqing's 2026-08-18 email: each match (Table 3) or tiebreak (Table 4) contributes **two** team-rows that are mechanically linked — a win for one team is a loss for the other, so the two rows are not independent draws. The main Tables 3–4 already address this via SE clustered by `match_id` (the project spec from the start). Two further checks were requested, both run here for both outcomes:

- **No-intercept spec**: keep both rows, drop the global intercept and give tournament×year FE full-rank (all-levels) dummies instead of the usual intercept + reference-dropped dummies. This spans the identical model as the baseline — same column space, same fitted probabilities — so it is a *verification*, not a new specification.
- **Random-one-team spec**: for each match (Table 3) / each specific tiebreak (Table 4), keep only one team's row via a coin flip, removing the mechanical complementarity entirely. Repeated over 10 random seeds, since any single draw is arbitrary. Robust (HC1) SE for Table 3 (exactly one row per match after de-duplication, so no clustering unit remains); SE stays clustered by `match_id` for Table 4 (a match can still contribute more than one tiebreak-row after de-duplication).

No-intercept check: exact reparameterization, as expected

The no-intercept spec reproduces Table 3 and Table 4 **exactly** — same AMEs, same SEs, same p-values, same log-likelihood to three decimals (e.g. same-nationality match-win LL = −2352.768 under both parameterizations). Dropping the intercept changes nothing here; it is confirmed to be a pure relabeling of the same fixed-effects structure, not an alternative model.

Random-one-team check: same direction, weaker power (as expected from halving N)

Outcome	Culture measure	Baseline AME	Random-one: mean AME (range, 10 seeds)	Significant (p<.05)
Match win (N=1,864 per draw)	Same nationality	+0.0361**	+0.0466 [+0.0161, +0.0812]	5/10
	Same language	+0.0459***	+0.0517 [+0.0230, +0.0877]	7/10
	Lang. proximity	+0.0419***	+0.0471 [+0.0156, +0.0859]	6/10
Tiebreak win (N≈1,177 per draw)	Same nationality	+0.0308 (n.s.)	+0.0174 [−0.0305, +0.0592]	1/10
	Same language	+0.0082 (n.s.)	−0.0041 [−0.0434, +0.0526]	0/10
	Lang. proximity	+0.0026 (n.s.)	−0.0107 [−0.0526, +0.0433]	0/10

All 10 tiebreak seeds converge in every spec (no singular-Hessian failures with the corrected, qualifying-round-excluded tiebreak panel).

Note on cross-checking against Stata: the no-intercept check above is a deterministic reparameterization and reproduces exactly in Stata (same AMEs, same LL to three decimals). The random-one-team draws, however, use each engine's own RNG (Stata's native generator vs. numpy's `default_rng`), so a given seed does not select the same rows in both — the specific mean AMEs, ranges, and per-seed significance counts reported here are Python-run numbers and are not expected to reproduce bit-for-bit in Stata. Re-running the check in Stata (2026-08-19) confirms the same qualitative pattern — match-win estimates stay positive-signed with reduced power, and the tiebreak result stays null throughout — but with different individual draws (e.g. Stata hit zero convergence failures across its 10 tiebreak seeds, and its tiebreak same-language mean AME came out slightly negative rather than slightly positive). Since the tiebreak baseline itself is a null, near-zero estimate, this sign wobble across engines is noise around zero, not a disagreement about the finding.

Every point estimate keeps the same sign and a broadly similar magnitude to its baseline — there is no evidence the two-rows-per-match design was manufacturing the effect. What changes is **power**: halving the sample inflates SEs by roughly √2, enough to push several draws below the 5% threshold — most visibly for same-nationality, already the weakest of the three culture measures at p=0.021 in the full sample, while same-language (p=0.002 at baseline, the strongest) stays significant in half the draws even with half the data. The tiebreak table's null result stays null either way. **Conclusion:** the stacked-both-rows + match-clustered design already used throughout this report is the correct and more efficient choice — the standard fix in the paired/matched-data literature — and the random-one-team check supports rather than overturns it.

5. Heterogeneity Analysis

Four sets of interactions are estimated on the main match-win sample (N = 3,728 team-obs), all with the full Table 3 control set (rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq): **Table 5**, culture × surface, testing surface-specific coordination demands; **Table 6**, culture × Hofstede individualism score (team-level, demeaned), testing whether collectivist teams gain more from cultural homophily; **Table 6a**, culture × years since turning pro (team average), testing whether experienced teams benefit more from cultural alignment; and **Table 6b**, culture × within-team experience gap, testing whether mismatched-experience teammates benefit more from cultural alignment.

Hofstede individualism scores (IDV) are taken from Geert Hofstede's official 6-dimension dataset (2015 release, geerthofstede.com). Countries not covered by Hofstede are assigned proxy scores following two tiers: *Tier 1* (direct cultural/linguistic kin, high confidence): BIH→SRB, CYP→GRC, MCO→FRA, MDA→ROU; *Tier 2* (same-region / closest available, moderate confidence): BLR→RUS, BOL→PER, DOM→COL, LBN→Arab composite (IDV=38), TUN→MAR, GEO→TUR, KAZ→RUS, UKR→RUS, UZB→RUS. The team-level score is IC_team = (IDV_p1 + IDV_p2) / 2, demeaned across the GS regression sample (mean = 62.98, SD = 20.85).

Table 5 — Heterogeneity: Culture × Surface | N = 3,728

Two specs, each pooling two surfaces into a single baseline rather than using all three at once. **Spec (a)**: grass vs. rest (hardcourt + clay pooled as baseline). **Spec (b)**: clay vs. rest (hardcourt + grass pooled as baseline). **No tournament×year FE** (round FE only) — with only two surface groups and no tournament×year FE, the surface main effect is directly estimable rather than absorbed, unlike the collinearity issue that arose with all three surfaces plus tournament×year FE together.

Variable	(1) Same Nationality		(2) Same Language		(3) Lang. Proximity	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Culture (C) — pooled baseline	+0.0395**	+0.0351*	+0.0439**	+0.0477***	+0.0416**	+0.0401**
	(0.0183) p=0.031	(0.0189) p=0.064	(0.0176) p=0.013	(0.0182) p=0.009	(0.0176) p=0.018	(0.0181) p=0.027
Surface (grass in (a) / clay in (b))	+0.0097	−0.0017	−0.0022	+0.0047	+0.0019	−0.0037
	(0.0149) p=0.515	(0.0142) p=0.903	(0.0207) p=0.916	(0.0188) p=0.804	(0.0209) p=0.927	(0.0194) p=0.850
C × Surface	−0.0169	+0.0023	+0.0070	−0.0084	+0.0003	+0.0061
	(0.0376) p=0.653	(0.0347) p=0.948	(0.0364) p=0.847	(0.0340) p=0.805	(0.0364) p=0.994	(0.0344) p=0.860
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0015	−0.0013	−0.0004	−0.0003	+0.0002	+0.0003
	(0.0304) p=0.962	(0.0304) p=0.965	(0.0304) p=0.990	(0.0305) p=0.992	(0.0304) p=0.994	(0.0305) p=0.992
Team avg. yrs since turning pro	+0.0044***	+0.0044***	+0.0042**	+0.0042**	+0.0042**	+0.0042**
	(0.0017) p=0.008	(0.0017) p=0.008	(0.0016) p=0.010	(0.0016) p=0.010	(0.0016) p=0.011	(0.0016) p=0.011
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.096	0.096	0.097	0.097	0.096	0.096
N		3,728		3,728		3,728

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: round only (no tournament×year FE). SE clustered by match. All surface main effects and interactions are insignificant ($p > 0.67$ throughout) in both specs. The culture advantage documented in Table 3 does not vary significantly by surface.

Table 5 robustness — tournament×year FE, interaction only (no surface main effect) | N = 3,728

Restores the full **tournament×year FE** (rather than dropping to round-only) and drops the separate surface main effect, keeping only Culture × grass and Culture × clay together in one spec (hardcourt is the implicit baseline for both interactions, all three surfaces entering through the interactions alone). This is the mirror image of the main Table 5 spec's tradeoff: T×Y FE is restored, but the surface main effect can no longer be separately identified from the tournament dummies it's nested in.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — hardcourt baseline	+0.0408*	+0.0467**	+0.0395*
	(0.0230) p=0.076	(0.0222) p=0.036	(0.0221) p=0.074
C × grass	−0.0181	+0.0042	+0.0023
	(0.0405) p=0.655	(0.0391) p=0.914	(0.0390) p=0.953
C × clay	−0.0026	−0.0069	+0.0071
	(0.0375) p=0.945	(0.0368) p=0.851	(0.0370) p=0.849
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0042	−0.0030	−0.0023
	(0.0315) p=0.894	(0.0315) p=0.925	(0.0315) p=0.941
Team avg. yrs since turning pro	+0.0044***	+0.0043**	+0.0042**
	(0.0017) p=0.008	(0.0017) p=0.010	(0.0017) p=0.011
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0008***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.096	0.097	0.097
N	3,728	3,728	3,728

Controls: same as Table 5 above. FE: tournament×year + round (T×Y FE restored). SE clustered by match. The culture main effect (evaluated at hardcourt) is somewhat larger and more significant here than in the round-FE spec (e.g. same nationality: +0.0395 → +0.0408; same language: +0.0439 → +0.0467), but both C×grass and C×clay interactions remain small and insignificant across all three culture measures (p > 0.65 throughout) — the same conclusion as the round-FE spec. The two specifications agree: the cultural homophily advantage does not vary meaningfully by surface, regardless of whether tournament×year FE or a surface main effect is used to absorb tournament-level variation.

Table 6 — Heterogeneity: Culture × Hofstede Individualism (demeaned) | N = 3,728

IC_team_dm = team-average Hofstede IDV score minus the sample mean (62.98). Positive values = more individualist than average. Hypothesis: more individualist teams rely less on in-group bonding → negative θ. Spec 2 only (includes the main IC effect) — the reported specification.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0293*	+0.0393**	+0.0352**
	(0.0167) p=0.079	(0.0163) p=0.016	(0.0163) p=0.031
IC_team_dm	+0.0011*	+0.0007	+0.0007
	(0.0006) p=0.065	(0.0008) p=0.356	(0.0008) p=0.349
C × IC_team_dm (θ)	−0.00046	+0.00007	+0.00009
	(0.00079) p=0.562	(0.00089) p=0.936	(0.00089) p=0.922
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0028	−0.0016	−0.0010
	(0.0316) p=0.930	(0.0316) p=0.960	(0.0316) p=0.975
Team avg. yrs since turning pro	+0.0051***	+0.0050***	+0.0049***
	(0.0017) p=0.003	(0.0017) p=0.004	(0.0017) p=0.004
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.002	(0.0002) p=0.002	(0.0002) p=0.002
Pseudo-R^2	0.097	0.098	0.097
N	3,728	3,728	3,728

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq. FE: tournament×year + round. SE clustered by match. The culture main effects are consistent with Table 3. The interaction θ is small and insignificant across all three measures (p > 0.56), providing no evidence that individualism moderates the cultural homophily advantage. The main IC_team_dm coefficient is positive and marginally significant for same nationality (p=0.065) — teams with higher individualism scores perform somewhat better overall — but is insignificant for the language measures (p>0.34).

Table 6 robustness — dropping Tier-2 Hofstede-proxy countries | N = 3,577

Team-obs involving a player whose IDV score relies on a Tier-2 (moderate-confidence) proxy — BLR, BOL, DOM, GEO, KAZ, LBN, TUN, UKR, UZB — are dropped (151 of 3,728 team-obs), leaving only direct Hofstede matches and Tier-1 (high-confidence) proxies.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	+0.0206	+0.0308*	+0.0266
	(0.0171) p=0.229	(0.0167) p=0.065	(0.0166) p=0.109
IC_team_dm	+0.0007	−0.0000	+0.0000
	(0.0006) p=0.264	(0.0008) p=0.999	(0.0008) p=0.998
C × IC_team_dm (θ)	+0.00009	+0.00090	+0.00092
	(0.00081) p=0.913	(0.00093) p=0.334	(0.00093) p=0.322
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0008***	+0.0008***	+0.0008***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0118	−0.0101	−0.0096
	(0.0329) p=0.720	(0.0330) p=0.759	(0.0330) p=0.770
Team avg. yrs since turning pro	+0.0053***	+0.0051***	+0.0051***
	(0.0017) p=0.002	(0.0017) p=0.003	(0.0017) p=0.003
Yrs since turning pro, squared	−0.0007***	−0.0007***	−0.0007***
	(0.0002) p=0.002	(0.0002) p=0.002	(0.0002) p=0.002
Pseudo-R^2	0.097	0.098	0.097
N	3,577	3,577	3,577

Same controls and FE as Table 6 above (exp_mean_dm carries the full-sample demeaning constant into this subsample rather than being re-centered), estimated on the Tier-2-excluded subsample. Point estimates on C shrink somewhat relative to the full-sample spec (e.g. same nationality: +0.0293 → +0.0206, no longer significant), consistent with the smaller, more homogeneous sample rather than a change in sign or a qualitatively different story. The IC_team_dm main effect for same nationality also loses significance (p=0.065 → p=0.264) once the weaker-confidence proxy countries are dropped, so that result should be treated as sensitive to the Tier-2 approximations. The θ interaction remains small and insignificant throughout, matching the full-sample conclusion that individualism does not moderate the cultural homophily advantage.

Table 6a — Heterogeneity: Culture × Years Since Turning Pro | N = 3,728

exp_mean is demeaned (exp_mean_dm = exp_mean − sample mean of 11.93 years since turning pro, SD = 5.17), consistent with how ic_team_dm is demeaned for Table 6. The main effect of culture (C) is therefore evaluated at the sample-average tenure level, not at zero; the interaction captures how the marginal effect changes per additional year since turning pro relative to that average. exp_mean_dm_sq = exp_mean_dm² (the demeaned quadratic control, used consistently across every table in this report) is included alongside exp_mean_dm, per the blanket rule that all heterogeneity tables carry the full Table 3/4 control set.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean experience	+0.0361**	+0.0433***	+0.0392**
	(0.0164) p=0.028	(0.0157) p=0.006	(0.0156) p=0.012
C × exp_mean_dm	+0.0008	+0.0069**	+0.0068**
	(0.0033) p=0.809	(0.0032) p=0.032	(0.0032) p=0.034
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0045	−0.0060	−0.0054
	(0.0315) p=0.886	(0.0313) p=0.848	(0.0313) p=0.864
Team avg. yrs since turning pro	+0.0041*	+0.0002	+0.0002
	(0.0023) p=0.070	(0.0025) p=0.944	(0.0025) p=0.948
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0008***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R²	0.096	0.098	0.097
N	3,728	3,728	3,728

Controls: rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq (full Table 3 control set). FE: tournament×year + round. SE clustered by match. With exp_mean_dm_sq back in the model, the culture main effects at mean experience are close to Table 3's own estimates (+0.0361/+0.0459/+0.0419) and all significant. The positive and significant interaction for language/proximity (p≈0.032–0.034) implies that advantage grows further with experience above the mean; nationality homophily shows no significant experience gradient (p=0.809).

Table 6a robustness — adding Culture × exp_mean_dm² | N = 3,728

Adds a quadratic interaction term, Culture × exp_mean_dm_sq, alongside the linear Culture × exp_mean_dm interaction reported above, to check the linear-interaction finding isn't an artefact of a curved true relationship. Same controls, FE, and clustering as the main Table 6a spec.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean experience	+0.0432**	+0.0357*	+0.0319
	(0.0197) p=0.028	(0.0195) p=0.066	(0.0194) p=0.100
C × exp_mean_dm	+0.0007	+0.0070**	+0.0069**
	(0.0033) p=0.839	(0.0032) p=0.028	(0.0032) p=0.031
C × exp_mean_dm² (quadratic)	−0.0003	+0.0003	+0.0003
	(0.0005) p=0.541	(0.0005) p=0.528	(0.0005) p=0.537
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0059	−0.0049	−0.0043
	(0.0315) p=0.852	(0.0314) p=0.876	(0.0314) p=0.890
Team avg. yrs since turning pro	+0.0040*	+0.0000	+0.0000
	(0.0022) p=0.072	(0.0025) p=0.986	(0.0025) p=0.988
Yrs since turning pro, squared	−0.0006*	−0.0010**	−0.0009**
	(0.0003) p=0.068	(0.0004) p=0.012	(0.0004) p=0.013
Pseudo-R^2	0.096	0.098	0.097
N	3,728	3,728	3,728

The quadratic interaction term is small and insignificant for all three culture measures ($p > 0.5$ throughout), so the curved-relationship concern doesn't hold up — the linear interaction finding from the main Table 6a spec stands. The linear C × exp_mean_dm term is essentially unchanged in sign, magnitude, and significance (language/proximity still positive and significant at $p \approx 0.03$; nationality still flat), confirming the main spec is not mis-specified by omitting the square.

Table 6b — Heterogeneity: Culture × Within-Team Experience Gap | N = 3,728

Tests a different hypothesis from Table 6a: rather than whether more *experienced* teams benefit more from cultural alignment, this asks whether *mismatched* teams do — the idea being that shared culture/language substitutes for shared on-court experience when one teammate is a veteran and the other is new. exp_gap = |exp_i − exp_j|, the absolute difference in each teammate's own years since turning pro (not the team average used elsewhere). Individual player tenure is imputed to 1 year for rookies (raw tenure ≤0), matching the exp_mean imputation rule; a team of two rookies has exp_gap = 0. Demeaned (exp_gap_dm = exp_gap − sample mean of 5.50 years, SD = 4.92) for the same reason every other interaction table is demeaned — so the culture main effect is evaluated at the average gap, not at a perfectly-matched (zero-gap) team. Controls: full Table 3 set (rank_mean, opp_rank_mean, single_top100, exp_mean_dm, exp_mean_dm_sq — team-average tenure and its square, same as every other table) plus exp_gap_dm itself as an added control, per the request that the gap variable be included alongside its interaction. FE: tournament×year + round. SE clustered by match.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean gap	+0.0356**	+0.0447***	+0.0409***
	(0.0164) p=0.030	(0.0157) p=0.004	(0.0157) p=0.009
C × exp_gap_dm	+0.0032	+0.0038	+0.0036
	(0.0034) p=0.344	(0.0032) p=0.232	(0.0032) p=0.261
exp_gap_dm (main)	−0.0036*	−0.0045*	−0.0045*
	(0.0021) p=0.082	(0.0025) p=0.065	(0.0025) p=0.067
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0075	−0.0063	−0.0055
	(0.0314) p=0.810	(0.0314) p=0.842	(0.0314) p=0.861
Team avg. yrs since turning pro	+0.0049***	+0.0051***	+0.0050***
	(0.0018) p=0.006	(0.0018) p=0.004	(0.0018) p=0.004
Yrs since turning pro, squared	−0.0008***	−0.0008***	−0.0008***
	(0.0002) p=0.001	(0.0002) p=0.001	(0.0002) p=0.001
Pseudo-R^2	0.097	0.097	0.097
N	3,728	3,728	3,728

The interaction is positive throughout — pointing in the direction the "communication substitutes for experience mismatch" hypothesis predicts — but is not statistically significant for any of the three culture measures (p = 0.23–0.34). The experience-gap main effect itself is negative and marginally significant (p≈0.07–0.08): teams with a bigger tenure mismatch between teammates do somewhat worse on average, independent of culture. So while the direction is consistent with the hypothesis, there isn't enough statistical power here to say cultural alignment specifically compensates for an experience mismatch — this is a different (and so far null) result from Table 6a's finding that cultural advantage grows with a team's own average experience.

Table 6b robustness — adding Culture × exp_gap_dm² | N = 3,728

Adds a quadratic interaction (Culture × exp_gap_dm²) and its corresponding control (exp_gap_dm²) alongside the linear terms above, per the same "add the squared interaction as a robustness check" request applied to Table 6a.

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) — mean gap	+0.0394**	+0.0416**	+0.0340*
	(0.0194) p=0.042	(0.0192) p=0.030	(0.0191) p=0.074
C × exp_gap_dm	+0.0039	+0.0030	+0.0019
	(0.0044) p=0.372	(0.0043) p=0.485	(0.0043) p=0.658
C × exp_gap_dm² (quadratic)	−0.0001	+0.0001	+0.0003
	(0.0005) p=0.790	(0.0005) p=0.763	(0.0005) p=0.522
exp_gap_dm (main)	−0.0030	−0.0032	−0.0026
	(0.0029) p=0.293	(0.0035) p=0.351	(0.0035) p=0.453
exp_gap_dm² (quadratic control)	−0.0001	−0.0002	−0.0003
	(0.0004) p=0.787	(0.0004) p=0.584	(0.0004) p=0.414
Team avg. doubles rank	−0.0007***	−0.0007***	−0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Opponent avg. doubles rank	+0.0007***	+0.0007***	+0.0007***
	(0.0001) p<0.001	(0.0001) p<0.001	(0.0001) p<0.001
Top-100 singles player	−0.0072	−0.0058	−0.0050
	(0.0314) p=0.820	(0.0314) p=0.855	(0.0314) p=0.874
Team avg. yrs since turning pro	+0.0052***	+0.0053***	+0.0052***
	(0.0019) p=0.006	(0.0018) p=0.004	(0.0018) p=0.005
Yrs since turning pro, squared	−0.0007**	−0.0007**	−0.0007**
	(0.0003) p=0.019	(0.0003) p=0.018	(0.0003) p=0.019
Pseudo-R²	0.097	0.098	0.097
N	3,728	3,728	3,728

The quadratic interaction and quadratic gap control are both small and insignificant throughout (p > 0.55), and the linear interaction stays positive but insignificant, matching the main spec's conclusion — no curved relationship being masked, and no evidence that cultural homophily specifically compensates for an experience mismatch within a team.

6. Partner Selection: Testing the Sorting Concern

The regressions above estimate the effect of cultural homophily on *performance*, conditional on a partnership having already formed. A separate concern is *selection*: if stronger players have more choice over who they partner with, and if they systematically choose culturally similar partners, that alone could produce the observed same-culture win advantage without cultural fit doing anything causal on court. This section tests whether partner selection is more assortative by culture among stronger players than among the field as a whole.

Unit of analysis: one row per *distinct partnership* — the same two players advancing through multiple rounds of one tournament are counted once, not once per round, since the partner choice was made at the start of the event (2,089 partnerships with complete ranking data across the four Grand Slams, 2018–2025, Olympics excluded). Doubles ranking is each player's own ranking as of that tournament (the same source `rank_mean` is built from). **Random-matching benchmark**: for each tournament-year, the closed-form expected rate under random pairing of that event's actual field, Σ_{pairs} 1[same country] / C(n,2) — and the parallel calculation for same-language and ling_prox, using the same CEPII `comlang_off`/`comlang_ethno` country-pair lookup that `same_language`/`ling_prox` are themselves built from (so the benchmark is on the identical scale as the actual rate, not a different measure). This is a closed-form expectation, not a simulation — deterministic and exactly reproducible. Missing country pairs not covered by the CEPII table (mostly very small nations) are imputed to 0 (not same-language / zero proximity) in the benchmark, a conservative choice that if anything pushes the benchmark rate *down* — so it does not overstate the actual-vs-random gap reported below.

Updated 2026-09-03: the 2,089-partnership / 4,178-ego-row counts below reflect the qualifying-round exclusion (Wimbledon 2018's 12 qualifying-round matches, present in the raw source but not part of the main draw, are now dropped upstream — see the Observation Breakdown note above). Re-deriving from the raw source (`men_matches_with_ranks_cleaned.xlsx`, sheet `players_list`) after that exclusion gives 2,096 distinct partnership-events, 2,089 after dropping incomplete-rank pairs, 4,178 ego-rows — matching the notebook and this table exactly.

Measure		All ranked players (N=2,089)	Both partners top-100 (N=1,211)	Both partners top-50 (N=712)
Same nationality	Actual	42.2%	32.6%	35.5%
	Random benchmark	5.2%	4.1%	4.5%
Same language	Actual	56.0%	49.7%	56.3%
	Random benchmark	16.2%	15.5%	15.8%
Ling. proximity	Actual	56.8%	50.6%	56.5%
	Random benchmark	19.9%	18.9%	18.9%

Two things stand out. First, **actual partner selection is far more culturally assortative than chance at every skill level** — same-nationality rates run 6–8× the random-matching benchmark, same-language rates 3–4×, throughout. Sorting by culture is real and substantial; Lingqing's concern that this could contaminate the performance regressions is well-founded and worth keeping in mind when interpreting Tables 3–6b as effects of culture on winning, rather than pure causal estimates. Second, and less expected: **elite players are not more assortative than the field as a whole** — the actual/benchmark ratio for same nationality is ≈8.1× overall, ≈7.9× for top-100, ≈7.9× for top-50, essentially flat across brackets (same-language ratio is similarly flat: ≈3.5×, ≈3.2×, ≈3.6×). If anything, the raw same-nationality rate is *lower* among top-50/100 pairs (32.6–35.5%) than in the full field (42.2%) — top players draw from a smaller, more international pool of plausible partners, which mechanically dilutes same-country pairing even though same-language pairing holds roughly steady. This doesn't support the specific "stronger players sort more" version of the selection concern — the sorting appears to be a broad feature of how doubles pairs form at every level, not something concentrated among top-ranked players with more partner choice.

6.1 Direct test: does a player's own ranking predict partner-culture similarity?

The bracket comparison above is a threshold test (top-50, top-100 vs. the field); this tests the same mechanism continuously. Each of the two players in a partnership is treated as an "ego" with the other as "partner" (2 ego-rows per partnership, 4,178 rows from the 2,089 partnerships), and the ego's **own** doubles ranking (at that tournament) is regressed on whether the partnership is same-nationality / same-language / ling_prox (all three binary; logit AME), with tournament×year FE and SE clustered by partnership (so both ego-rows from the same team don't count as independent observations). The model is deliberately bivariate — `own_rank + C(tourn_year)`, no further controls — since the point is to test whether own ranking alone predicts partner-culture similarity net of which event/year (which differ in nationality mix), not to explain partner-culture similarity fully; several of the Tables-3–6b controls don't have a meaningful analogue here (e.g. `opp_rank_mean` presumes a match opponent, which doesn't exist for a partner-formation outcome). The table below therefore already shows the complete specification, not an abridged one.

Outcome	Effect of own_rank	SE	p	N (ego-rows)
Same nationality (AME)	+0.000268***	0.000056	<0.001	4,178
Same language (AME)	+0.000145**	0.000058	0.013	4,178
Ling. proximity (OLS coef.)	+0.000098**	0.000044	0.026	4,178

Ling. proximity's row above uses an OLS coefficient, not an AME, since `ling_prox` (`prox1`, continuous) is not a binary outcome — logit AME does not apply.

Doubles ranking runs the opposite direction from skill (rank 1 = best), so a **positive** coefficient means a *worse*-ranked player is *more* likely to have a culturally similar partner — the reverse of what the "stronger players sort more" concern predicts. All three outcomes point the same way, and each is statistically significant. This is the same conclusion as the bracket comparison in the table above, now shown as a continuous, statistically-tested relationship rather than a three-bucket comparison: the data give no support for the mechanism that stronger players' greater partner choice is what generates the observed cultural homophily. If anything, weaker-ranked players sort into same-culture partnerships slightly more, plausibly because players with less market power to attract a foreign/elite partner default to whoever is available in their own federation or language community.

6.2 Extended specification: field composition

Section 6.1's `own_rank`-only spec doesn't account for the cultural composition of the tournament field a player is drawn from, which varies by tournament-year and could mechanically affect how likely any given player is to end up with a culturally similar partner. We estimate three specifications, all on the same singleton-nationality-excluded sample (below): **(1)** own_rank alone (Section 6.1, reproduced below); **(2)** the outcome-matched field-composition variable alone; **(3)** own_rank plus field composition together. Field composition is computed from the *full* tournament-year field (every entrant, not only the rank-complete partnerships used above): for `same_country`, the share of every *other* field player sharing the focal player's nationality, (same-nat field count − 1) / (field size − 1); for `same_language`/`ling_prox`, the analogous mean pairwise value against every other field player, via the same CEPII lookup used throughout. All three specs also exclude the 3 ego-rows whose own nationality is represented by exactly one player in the ego-row sample (so a single lone-nationality observation cannot by itself drive any coefficient), leaving N=4,175 (from 4,178).

Outcome	Variable	Spec 1	Spec 2	Spec 3
Same nationality (AME)	own_rank	+0.000269***		+0.000116**
		(0.000056)		(0.000053)
	composition		+3.853***	+3.756***
			(0.241325)	(0.255403)
Same language (AME)	own_rank	+0.000146**		+0.000117**
		(0.000058)		(0.000053)
	composition		+1.275***	+1.267***
			(0.081709)	(0.082578)
Ling. proximity (OLS coef.)	own_rank	+0.000098**		+0.000088**
		(0.000044)		(0.000038)
	composition		+1.249***	+1.247***
			(0.116322)	(0.117364)
N (ego-rows)		4,175	4,175	4,175

Ling. proximity's row uses OLS coefficients throughout, not AME, since `ling_prox` (`prox1`, continuous) is not a binary outcome.

Field composition alone (Spec 2) is a strong, highly significant predictor of all three outcomes, as expected — a player drawn from a field with more compatriots or linguistically similar entrants is mechanically more likely to end up with a culturally similar partner. Adding both own ranking and field composition together (Spec 3) shrinks the own-ranking coefficients by roughly half relative to Spec 1 but leaves them positive and significant for all three outcomes ($p<0.05$ throughout), while the composition coefficient stays close to its Spec 2 magnitude. Own ranking therefore predicts partner similarity *in addition to*, not merely *because of*, the cultural composition of the field a player happens to be drawn from.